

Recommendation

ITU-T Y.3804 (04/2025)

SERIES Y: Global information infrastructure, Internet protocol aspects, next-generation networks, Internet of Things and smart cities

Quantum key distribution networks

Quantum key distribution networks – Control and management

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Recommendation ITU-T Y.3804

Quantum key distribution networks – Control and management

Summary

To realize secure, stable, efficient, and robust operations of and services by a quantum key distribution (QKD) network as well as to manage a QKD network (QKDN) as a whole and support user network management, Recommendation ITU-T Y.3804 specifies functions and procedures for QKDN control and management based on the requirements specified in Recommendation ITU-T Y.3801.

This revision of the Recommendation adds the description of reference point Mx and the management procedures in Subscription-Notification information exchange.

History*

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Recommendation ITU-T Y.3804

Quantum key distribution networks – Control and management

1 Scope

This Recommendation specifies control and management functions and procedures for quantum key distribution (QKD) networks. This Recommendation covers:

- Functional elements of QKD network (QKDN) control, management, and orchestration;
- Functions of QKDN control, management, and orchestration;
- Procedures of QKDN control, management, and orchestration.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

- [[ITU-T M.3400](#)] Recommendation ITU-T M.3400 (2000), *TMN management functions*.
- [[ITU-T Y.3111](#)] Recommendation ITU-T Y.3111 (2017), *IMT-2020 network management and orchestration framework*.
- [[ITU-T Y.3800](#)] Recommendation ITU-T Y.3800 (2019), *Overview on networks supporting quantum key distribution*.
- [[ITU-T Y.3801](#)] Recommendation ITU-T Y.3801 (2020), *Functional requirements for quantum key distribution networks*.
- [[ITU-T Y.3802](#)] Recommendation ITU-T Y.3802 (2020), *Quantum key distribution networks – Functional architecture*.
- [[ITU-T Y.3803](#)] Recommendation ITU-T Y.3803 (2020), *Quantum key distribution networks – Key management*.

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 key data [ITU-T Y.3803]: Random bit strings, which are used as a cryptographic key.

3.1.2 key life cycle [ITU-T Y.3800]: A sequence of steps that a key undergoes from its reception by a key manager (KM) through its use in a cryptographic application and until deletion or preservation depending on the key management policy.

3.1.3 key management [ITU-T Y.3800]: All activities performed on keys during their life cycle starting from their reception from the quantum layer, storage, formatting, relay, synchronization, authentication, to supply to a cryptographic application and deletion or preservation depending on the key management policy.

3.1.4 key management agent (KMA) [ITU-T Y.3802]: A functional element to manage keys generated by one or multiple quantum key distribution (QKD) modules in a QKD node (trusted node).

NOTE – KMA acquires keys from one or multiple QKD modules, synchronizes, resizes, formats, and stores them. It also relays keys KMA links.

3.1.5 key management agent-key (KMA-key) [ITU-T Y.3803]: Key data stored and processed in a key management agent (KMA) and securely shared between a KMA and a matching KMA.

3.1.6 key manager (KM) [ITU-T Y.3800]: A functional module located in a quantum key distribution (QKD) node to perform key management in the key management layer.

3.1.7 key manager link [ITU-T Y.3800]: A communication link connecting key managers (KMs) to perform key management.

3.1.8 key relay [ITU-T Y.3800]: A method to share keys between arbitrary quantum key distribution (QKD) nodes via intermediate QKD node(s).

3.1.9 key supply [ITU-T Y.3800]: A function providing keys to cryptographic applications.

3.1.10 key supply agent (KSA) [ITU-T Y.3802]: A functional element to supply keys to a cryptographic application, being located between a key management agent (KMA) and the cryptographic application.

NOTE – Application interfaces for cryptographic applications are installed into the key supply agent (KSA). The KSA synchronizes keys and verifies their integrity via a KSA link before supplying them to the cryptographic application.

3.1.11 key supply agent-key (KSA-key) [ITU-T Y.3803]: Key data stored and processed in a key supply agent (KSA), and securely shared between a KSA and a matching KSA.

3.1.12 quantum key distribution module (QKD module) [ITU-T Y.3800]: A set of hardware and software components that implements cryptographic functions and quantum optical processes, including quantum key distribution (QKD) protocols, synchronization, distillation for key generation, and is contained within a defined cryptographic boundary.

NOTE – A QKD module is connected to a QKD link, acting as an endpoint module in which a key is generated. These are two types of QKD modules, namely, the transmitters (QKD-Tx) and the receivers (QKD-Rx).

3.1.13 quantum key distribution link (QKD link) [ITU-T Y.3800]: A communication link between two quantum key distribution (QKD) modules to operate the QKD.

NOTE – A QKD link consists of a quantum channel for the transmission of quantum signals, and a classical channel used to exchange information for synchronization and key distillation.

3.1.14 quantum key distribution network (QKDN) [ITU-T Y.3800]: A network comprised of two or more quantum key distribution (QKD) nodes connected through QKD links.

NOTE – A QKDN allows sharing keys between the QKD nodes by key relay when they are not directly connected by a QKD link.

3.1.15 quantum key distribution network controller (QKDN controller) [ITU-T Y.3800]: A functional module, which is located in a quantum key distribution (QKD) network control layer to control a QKD network.

3.1.16 quantum key distribution network manager (QKDN manager) [ITU-T Y.3800]: A functional module, which is located in a quantum key distribution (QKD) network management layer to monitor and manage a QKD network.

3.1.17 quantum key distribution node (QKD node) [ITU-T Y.3800]: A node that contains one or more quantum key distribution (QKD) modules protected against intrusion and attacks by unauthorized parties.

NOTE – A QKD node can contain a key manager (KM).

3.1.18 user network [ITU-T Y.3800]: A network in which cryptographic applications consume keys supplied by a quantum key distribution (QKD) network.

NOTE – In this Recommendation, "key" means "symmetric random bit strings" produced by QKDN.

3.2 Terms defined in this Recommendation

None.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

BSS	Business Support System
CDR	Charging Data Record
FCAPS	Fault, Configuration, Accounting, Performance and Security
ID	Identifier
IMT	International Mobile Telecommunications
IP	Internet Protocol
KM	Key Manager
KMA	Key Management Agent
KMLM	Key Management Layer Management
KSA	Key Supply Agent
OSS	Operation Support System
QBER	Quantum Bit Error Rate
QCLM	QKDN Control Layer Management
QKD	Quantum Key Distribution
QKDN	Quantum Key Distribution Network
QLM	Quantum Layer Management
QoS	Quality of Service
TMN	Telecommunications Management Network
VPN	Virtual Private Network
XLMO	Cross Layer Management and Orchestration

5 Conventions

None.

6 Overview

This Recommendation specifies functional elements and associated functions, reference points, and procedures for the QKDN control and management. Basic control and management functions and layered structures of QKDN are defined in [ITU-T Y.3800]. More specific functions are:

- control and management specific functions (e.g., path computation for routing control, session control including access traffic steering/switching/splitting for session control, quality of service (QoS) and charging policy control, FCAPS management for each layer);

- control and management reference points among/between control and management functional components and those of other layers;
- control and management orchestration functions of multi-layers. QKDN management layer includes multiple functional components responsible for multi-layers (quantum, key management, and QKDN control layers) and cross-layer management orchestration;
- interworking functions with external management systems especially user network management system, management capability exposure function, etc.

7 Functional architecture of the control, management, and orchestration of QKDN

This clause specifies functional components of QKDN control and management. Figure 1 highlights functional components and reference points relevant to QKDN control and management in a QKDN. Each layer has a layer specific control and management function associated with a corresponding management function in the QKDN management layer. Each layer specific control and management function provides a management agent capability between each layer management function of the QKDN manager and its respective layer functions. Cross-layer management orchestration function provides orchestration capability among multiple layer management functions. Reference points Cx, Ck, Cq, Cops and Cqrp are defined as standard interfaces between the QKDN controller(s) and the functional components under control for the purpose of QKDN control.

Reference points Mq, Mqrp, Mops, Mk, Mc, Mx and Mu are defined as standard interfaces between the QKDN manager and the functional components under management for the purpose of QKDN management.

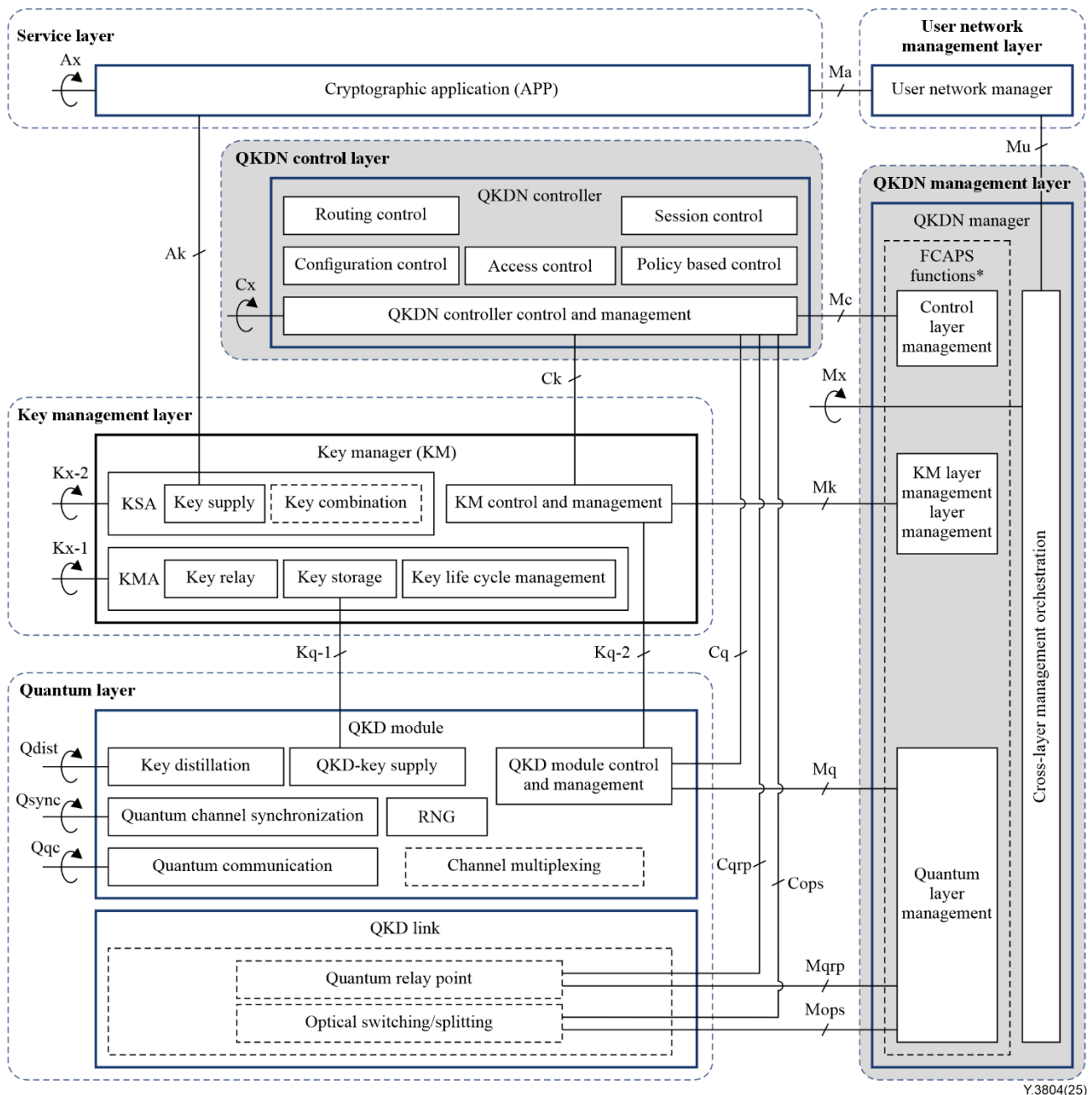


Figure 1 – Functional elements and reference points relevant to QKDN control and management

8 QKDN control layer

In order to realize secure, stable, efficient, and robust operations of and services by a QKDN, the QKDN controller(s) is/are introduced. The QKDN controller controls the quantum layer, the key management layer, and supports the functions of the QKDN management layer and the service layer.

The QKDN controller communicates control information with the KMs, the QKD modules, and the QKD link via the reference points Ck, Cq, Cqrp, and Cops, respectively. In the distributed architecture, the QKD controllers communicate with each other via the reference point Cx.

The QKDN controller communicates management information with the QKDN manager via the reference point Mc as recommended by Req_C 7 in [ITU-T Y.3801]. This function supports the FCAPS of QKDN.

The QKDN controller does not deal with keys themselves. Keys are supplied directly from a KM to a cryptographic application via the reference point Ak.

The QKDN controller performs the following functions:

- Routing control;
- Configuration control;
- Policy-based control;
- Access control;
- Session control.

8.1 Routing control function

As required by Req_C 1 in [ITU-T Y.3801], the routing control function provisions an appropriate key relay route between two end points of KMs in the key management layer.

Routing control function also performs the rerouting of key relay depending on the status of the key management layer and/or the quantum layer, ensuring the continued operation of the key supply/key relay.

For these purposes, the QKDN controller has the following functionalities:

- To manage a routing table which contains the necessary information on QKD node addresses and KM IDs;
- To acquire information on key consumption rate and the residual amount of keys from the KMs, QKD link parameters from the QKD modules, and QKDN topology information from QKDN manager;
- To be supported on optimization of key relay routes in the entire QKDN by the QKDN manager, which monitors the whole status of the quantum layer and the key management layer, and registers and updates them in a database.

The QKDN controller performs the provisioning and rerouting of key relay routes in the following ways:

1) Provisioning of key relay route

The two end-point KMs inform the QKDN controller of the required amount of keys from the two end-point cryptographic applications.

The QKDN controller analyses the status of the key management layer, especially on key consumption rate and the residual number of keys of the relevant KMs together with likely candidates of key relay routes.

The QKDN controller then finds and provisions an appropriate key relay route.

2) Rerouting of key relay

There are two typical cases where the rerouting is carried out.

Case 1: The key management layer attributed

- The residual amount of keys in the KM(s) in the relay node(s) runs short below a threshold.
- A fault is detected in the KM(s) in the relay node(s) and the KM link(s) connected to the relay node(s).

Case 2: The quantum layer attributed

- A quantum bit error rate (QBER) increases beyond a threshold in a certain QKD link connected to the relay node(s).
- A fault is detected in the QKD module(s) in the relay node(s).

In any case, faulty KMs, KM links, QKD modules, and QKD link are recommended to be de-activated, and the appropriate cause analysis and counter-measures should be applied by the QKDN manager.

Rerouting methods may include fixed rate rerouting, data-traffic adaptive rerouting, and scheduled rerouting as described in clause 8.2 of [ITU-T Y.3803] "Management of KMA-key and KSA-key in the key management layer".

8.2 Configuration control function

As recommended by Req_C 2 in [ITU-T Y.3801], the configuration control performs the following functions:

- To acquire control related configuration information on QKD modules and QKD links in the quantum layer and KMs and KM links in the key management layer;
- To control the state of these components (in service, out of service, standby, or reserved);
- To reconfigure QKD links and KM links if an alarm or failure diagnosis is notified.

In particular, alarms and failure diagnosis on QKD links include the increase of QBER implying that eavesdropping is launched against a quantum channel, or the loss of quantum channel is increased. The reconfiguration of a QKD link is made by controlling optical switch/splitter, modules in the quantum relay points, or by bringing a backup quantum channel into service.

The reconfiguration of KM links is mainly to replace faulty classical channels with new ones, and bring them into service, which actually is the same as that of a conventional telecommunications network.

8.3 Policy-based control function

Policy-based control function controls the QKDN based on QoS, key management and charging policies for cryptographic applications. Policy decision making is performed with the help of the QKDN manager.

8.4 Access control function

As recommended by Req_C 5 in [ITU-T Y.3801], the access control function provides capabilities to verify the claimed identity of functional components under the control and support of the QKDN controller and to restrict the functional components to pre-authorized activities or roles. Verified identities and their authorized rights/roles provide the basis of secure operations of and services by the QKDN.

The access control function has an access control repository of QKD nodes, QKD modules, KMs, and cryptographic applications. The access control repository also contains authorized roles, access rights of these functional components, and their priorities based on enforced policies. Based on this repository, the access control performs the following functions:

- To register and delete IDs of functional components;
- To issue certificates to the registered functional components;
- To perform authentication between the QKDN controller and a functional component, referring to their certificates;
- To support authentication between the functional components according to their certificates and the access control repository.

Detailed security mechanisms of access control function are outside the scope of this Recommendation.

8.5 Session control function

As recommended by Req_C 6 in [ITU-T Y.3801], the session control function supports the KMAs to establish the end-to-end key and/or the key supply agents (KSAs) to supply keys to cryptographic applications in the service layer of the user network which are specified in [ITU-T Y.3802] and [ITU-T Y.3803]. The KMA controls the session based on key management policy and the KSA controls the session procedure of flows of key supply services for multiple cryptographic applications, based on charging policies enforced by the policy-based control function (see clause 8.3).

9 QKDN management layer

A QKDN management layer provides functions to manage a QKDN as a whole and support user network management. QKDN management functions are categorized into five functional areas, namely, FCAPS management just like those of other networks.

The generic functional aspects of FCAPS management have already been specified in detail, for example, by [ITU-T M.3400] for the telecommunications management network (TMN) and [ITU-T Y.3111] for the international mobile telecommunications (IMT)-2020. Some of those specifications can be directly imported into QKDN management function specifications, while some others have been already covered by functions of key management and QKDN control.

This Recommendation focuses on the FCAPS management aspects for QKDN.

9.1 Common management functions

QKDN specific aspects of FCAPS arise especially from configurational restriction on the quantum layer due to point-to-point nature of QKD, and from security concerns due to natures of key establishment infrastructure. Thus, these aspects are related mostly to the quantum layer and the key management layer. Taking these aspects into account, the common FCAPS management functions across quantum, key management, and QKDN control layers of the QKDN are as follows:

As required by requirement Req_M 1 in [ITU-T Y.3801], the QKDN manager supports the following fault management functions:

- The QKDN manager monitors QKD link failures to support QKD modules for appropriate recovery actions including reconfiguration of QKD links and rerouting of key relay routes.

As recommended by requirement Req_M 2 in [ITU-T Y.3801], the QKDN manager supports the following fault management functions:

- The QKDN manager provides fault detection and root-cause analysis/diagnosis capability for quantum, key management, and QKDN control layers;
- The QKDN manager makes decisions and generation of failure resolving policies and interacts with each layer for healing actions;
- The QKDN manager discovers each layer managed resources and functions and bootstrap to make them ready for the operation based on the bootstrapping policies.

As required by requirement Req_M 3 in [ITU-T Y.3801], the QKDN manager supports the following configuration management functions:

- The QKDN manager provisions and configures the managed resources in each layer.

As recommended by requirement Req_M 4 in [ITU-T Y.3801], the QKDN manager supports the following configuration management functions:

- The QKDN manager manages the configuration status of each layer;
- The QKDN manager manages the network topology of each layer;
- The QKDN manager performs inventory management for all the QKDN resources in each layer;

- The QKDN manager manages the life cycle of the resource repositories (e.g., create, store, retrieve, modify, remove, etc.) in each layer.

As recommended by requirement Req_M 5 in [ITU-T Y.3801], the QKDN manager supports the following accounting management functions:

- The QKDN manager measures the resource usage data of each layer (e.g., usage for quantum keys in quantum layer) and generates accounting policies for charging.

As required by Req_M 6 in [ITU-T Y.3801], the QKDN manager supports the following performance management functions:

- The QKDN manager collects the performance data and status of each layer, registers them into a performance database, and updates them;
- The QKDN manager analyses the performance of collected data and generates performance reports.

As recommended by Req_M 7 in [ITU-T Y.3801], the QKDN manager supports the following performance management functions:

- The QKDN manager manages the key supply service policies.

As required by requirements Req_M 8 and M 9 in [ITU-T Y.3801], the QKDN manager supports the following security management functions:

- The QKDN manager collects management information including metadata, event logs, audit trail, and so on from each layer for detecting security anomalies;
- The QKDN manager supports key life cycle management by KMs, ensuring traceability of keys, by using the log database.

As recommended by requirement Req_M 10 in [ITU-T Y.3801], the QKDN manager supports the following security management functions:

- The QKDN manager has a root certification authority which issues root certificates to the QKDN controller. The QKDN manager supports the QKDN controller for the access control;
- The QKDN manager manages the key management policies and transmits them to the QKDN controller.

9.2 Layer specific management functions

A QKDN manager provides FCAPS management functions for each QKDN layer: the quantum layer, the key management layer, and the QKDN control layer. Thus, the QKDN manager contains the following three-layer specific functions:

- Quantum layer management (QLM) functions;
- Key management layer management (KMLM) functions;
- QKDN control layer management (QCLM) functions.

The above three functions perform FCAPS management. Also, for coordination, cross-layer, and external management issues, the management functions of the three layers can be orchestrated by:

- Cross-layer management orchestration (XLMO) functions,

which also support user network management.

9.2.1 QLM functions

The quantum layer specific FCAPS management functions are as follows:

- The QKDN manager detects eavesdropping attempts against a quantum channel;
- The QKDN manager performs collection and analysis of QKD specific performance information such as key generation rates;

- The QKDN manager manages availability and reliability of quantum key distribution based on redundancy of QKD links provided by the quantum layer;
NOTE – Redundancy of QKD links for their availability and reliability is assumed to be a well-accepted practice by QKDN operators. Thus, the associated requirement does not explicitly need to be defined.
- The QKDN manager supports meta data abstraction to map device dependent data into device independent data for the device interoperability in the quantum layer.

9.2.2 KMLM functions

The key management layer specific FCAPS management functions are as follows:

- The QKDN manager performs collection and analysis of available amounts of keys in KM for key relay, key supply services and the key life cycle management.

9.2.3 QCLM functions

The QKDN control layer specific FCAPS management functions are as follows:

- The QKDN manager supports the QKDN controller for routing and rerouting of key relay including instruction of policies and rules caused by the faults or performance degradation;
- The QKDN manager supports the QKDN controller for provisioning of routing and re-routing of key relay routes if QKDN supports key relay as the configuration management function.

9.3 XLMO functions

XLMO orchestrates the quantum layer, key management layer, and QKDN control layer management functions. It also orchestrates control and management functions. It exchanges management information with external management entities. The user network management layer is an external management entity from the QKDN management point of view. It can also interact and coordinate with any other management entities (e.g., operator's OSS, BSS, etc.) if necessary.

As recommended by Req_M 11 in [ITU-T Y.3801], the XLMO supports the management functions as outlined in clauses 9.3.1 and 9.3.2.

9.3.1 Orchestration for cross-layers management

- XLMO provides management coordination of the quantum layer, key management layer, and QKDN control layer;
- XLMO provides management orchestration of the QKDN control layer and QKDN management layer to support the QKDN controller to take necessary actions for anomalous situations (e.g., fault, performance degradation, security attacks, etc.);
- For entire QKDN provisioning, XLMO divides the provisioning information into three different types of initialization and configuration information for three layers (i.e., quantum layer, key management layer and QKDN control layer) and performs provisioning tasks per layer in sequence.

9.3.2 Orchestration for external management

- XLMO provides management orchestration with external management systems, especially with the user network management system and between two QKDN managers in the same QKDN;
- XLMO collects, stores and displays the topology of the QKDN;
- XLMO provides the QKDN resource usage status;
- XLMO supports the QKDN resource provisioning requested by the user network manager;
- XLMO provides friendly interaction with the user network manager on behalf of the users.

10 Procedures of control, management and orchestration of QKDN

This clause includes basic operational procedures of the control, management, and orchestration of QKDN. The details in each procedure are assumed to be arranged and/or varied depending on implementation and service use cases of the QKDN. These varied procedures are also conformed with this Recommendation.

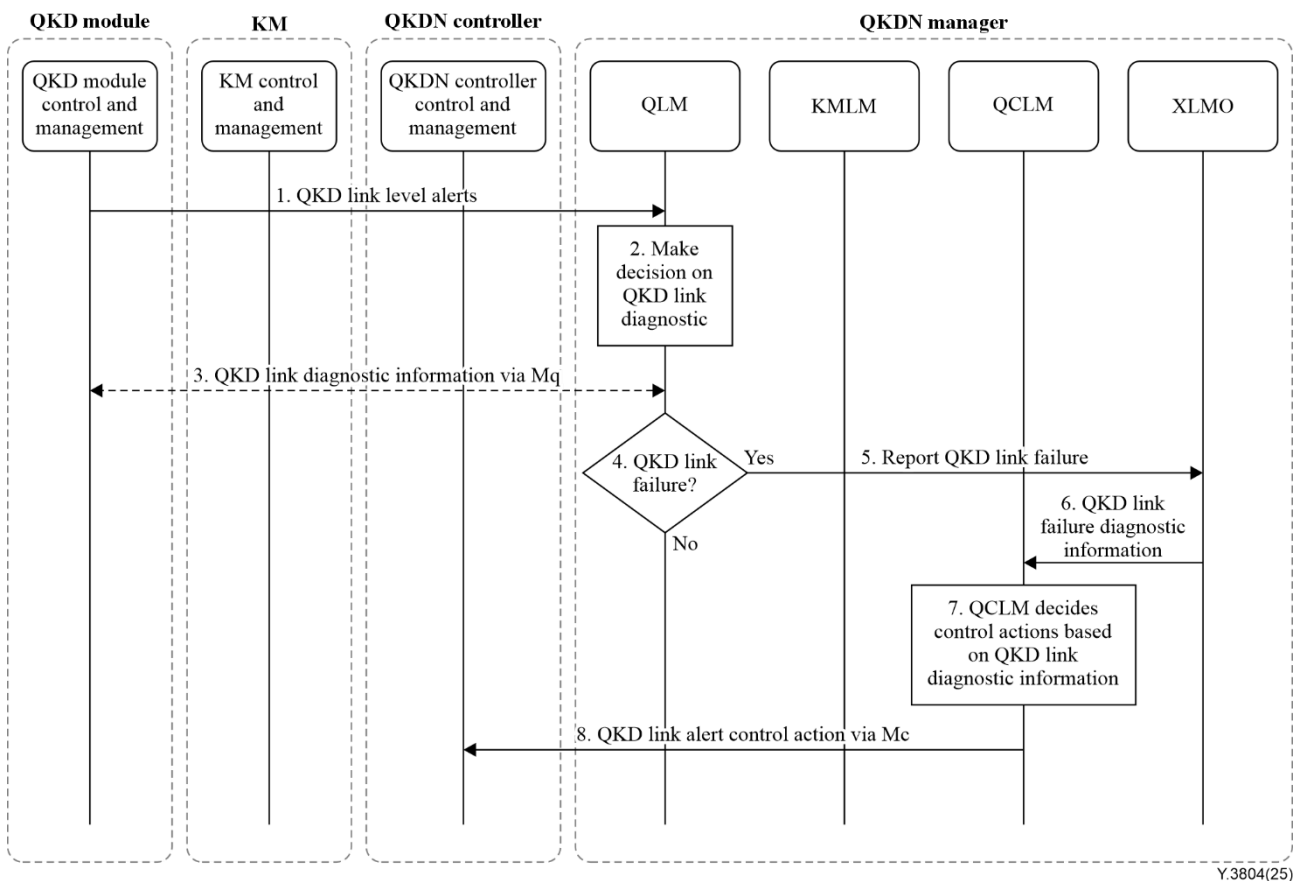
10.1 Fault management procedure

10.1.1 Fault management procedure: QKD link failure

(1) Request-response information exchange

A fault management procedure for QKD link failure is shown in Figure 2.

- 1) When a link failure occurs, a QLM receives a QKD link level alert.
- 2) The QLM decides the initiation of link diagnostic.
- 3) The QLM optionally sends a QKD link diagnostics request to the QKD module if necessary, and the QKD module reports additional QKD link diagnostic information.
- 4) The QLM analyses QKD link diagnostic information.
- 5) The QLM reports the QKD link status to the XLMO based on the QKD link diagnostic information.
- 6) The XLMO sends QKD link diagnostic information to the QCLM.
- 7) The QCLM makes appropriate control decision(s) to deal with the alert.
- 8) The QCLM sends it to the QKDN controller for the required actions.



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Figure 2 – A fault management procedure in request-response information exchange: QKD link failure

(2) Subscription-notification information exchange

A fault management procedure for QKD link failure is shown in Figure 3.

- 1) A QLM subscribes to data collection by invoking the DataCollectionSubscription_Subscribe operation to the QKD module control and management function via the Mq reference point. The QKD module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the QLM are listed in Table II.1. The TypeOperation indicates fault management in this step.

- 2) The QKD module control and management function notifies the QLM with data collection by invoking DataCollectionSubscription_Notify operation, based on the subscription in step 1).

NOTE 2 – For example, QKD link level alert is notified when a link failure occurs. The information components that can be provided to the QLM are listed in Table II.2. The DataCollectionSubscription_Notify optionally includes an EventNotify. The EventNotify can represent additional QKD link diagnostic information in this step. In this case, steps 3), 4) can be omitted.

- 3) The QLM decides the initiation of link diagnostic.
- 4) The QLM optionally sends a QKD link diagnostics request to a QKD module if necessary, and the QKD module reports additional QKD link diagnostic information.
- 5) The QLM analyses QKD link diagnostic information.
- 6) The QLM reports the QKD link status to the XLMO based on the QKD link diagnostic information.
- 7) The XLMO sends QKD link diagnostic information to the QCLM.
- 8) The QCLM makes appropriate control decision(s) to deal with the alert.
- 9) The QCLM sends it to the QKDN controller for the required actions.

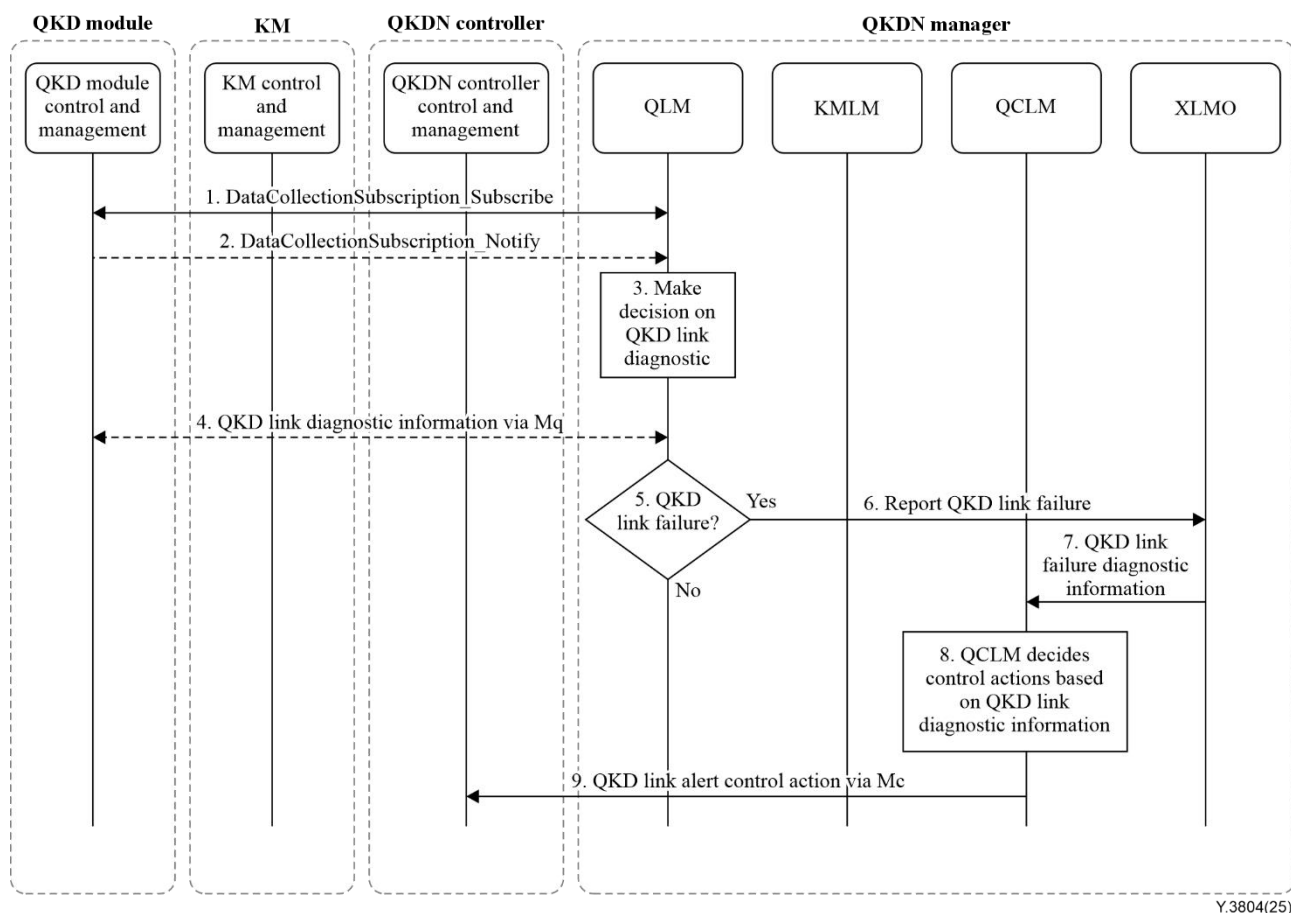


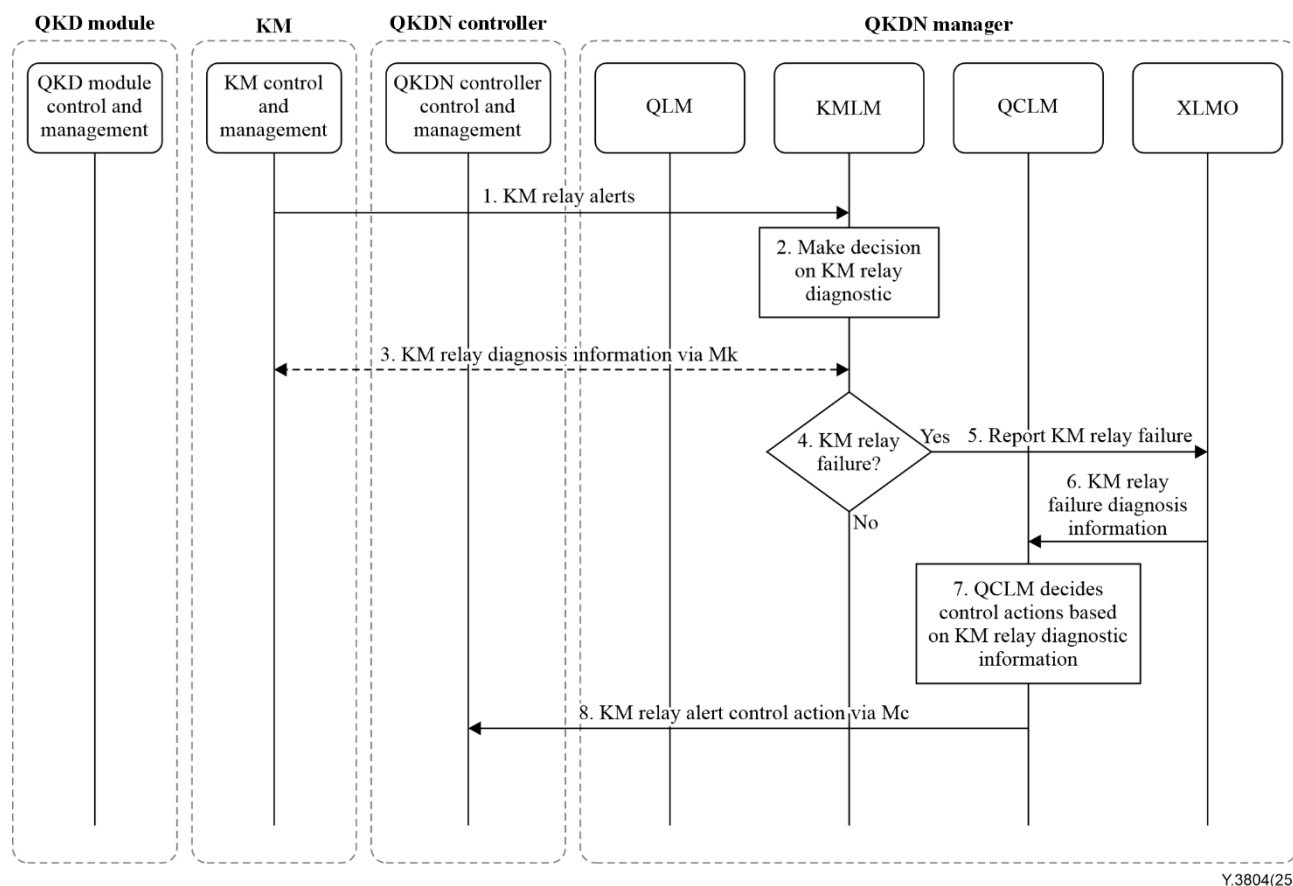
Figure 3 – A fault management procedure in subscription-notification information exchange: QKD link failure

10.1.2 Fault management procedure: key relay failure in KM

(1) Request-response information exchange

A management procedure for key relay failure in KM is shown in Figure 4.

- 1) When the KM failure occurs, a KMLM receives an alert such as KM relay alert.
- 2) The KMLM decides initiation of KM relay diagnostic.
- 3) The KMLM optionally sends KM relay diagnostics request to the KM and receives a report of KM relay diagnostic information if necessary.
- 4) The KMLM analyses KM relay diagnostic information and checks if the failure is related with key relay.
- 5) If yes, the KMLM reports the key relay failure in the KM to the XLMO based on the KM relay diagnostic information.
- 6) The XLMO sends it to the QCLM.
- 7) The QCLM makes appropriate control decision(s) to deal with the alert.
- 8) The QCLM sends it to the QKDN controller for the required actions.



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Figure 4 – A fault management procedure in request-response information exchange: Key relay failure in KM

(2) Subscription-notification information exchange

A fault management procedure for key relay failure in KM is shown in Figure 5.

- 1) A KMLM subscribes to data collection by invoking the `DataCollectionSubscription_Subscribe` operation to the KM module control and management function via the Mk reference point. The KM module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the KMLM are listed in Table II.1. The `TypeOperation` indicates fault management in this step.

- 2) The KM module control and management function notifies the KMLM with data collection by invoking `DataCollectionSubscription_Notify` operation, based on the subscription in step 1).

NOTE 2 – For example, KM relay alert is notified when a KM failure occurs. The information components that can be provided to the KMLM are listed in Table II.1. The `DataCollectionSubscription_Notify` optionally includes an `EventNotify`. The `EventNotify` can represent a report of KM relay diagnostic information in this step. In this case, step 3), 4) can be omitted.

- 3) The KMLM decides initiation of KM relay diagnostic.
- 4) The KMLM optionally sends KM relay diagnostics request to the KM and receives a report of KM relay diagnostic information if necessary.
- 5) The KMLM analyses KM relay diagnostic information and checks if the failure is related with key relay.
- 6) If yes, the KMLM reports the key relay failure in the KM to the XLMO based on the KM relay diagnostic information.
- 7) The XLMO sends it to the QCLM.

- 8) The QCLM makes appropriate control decision(s) to deal with the alert.
- 9) The QCLM sends it to the QKDN controller for the required actions.

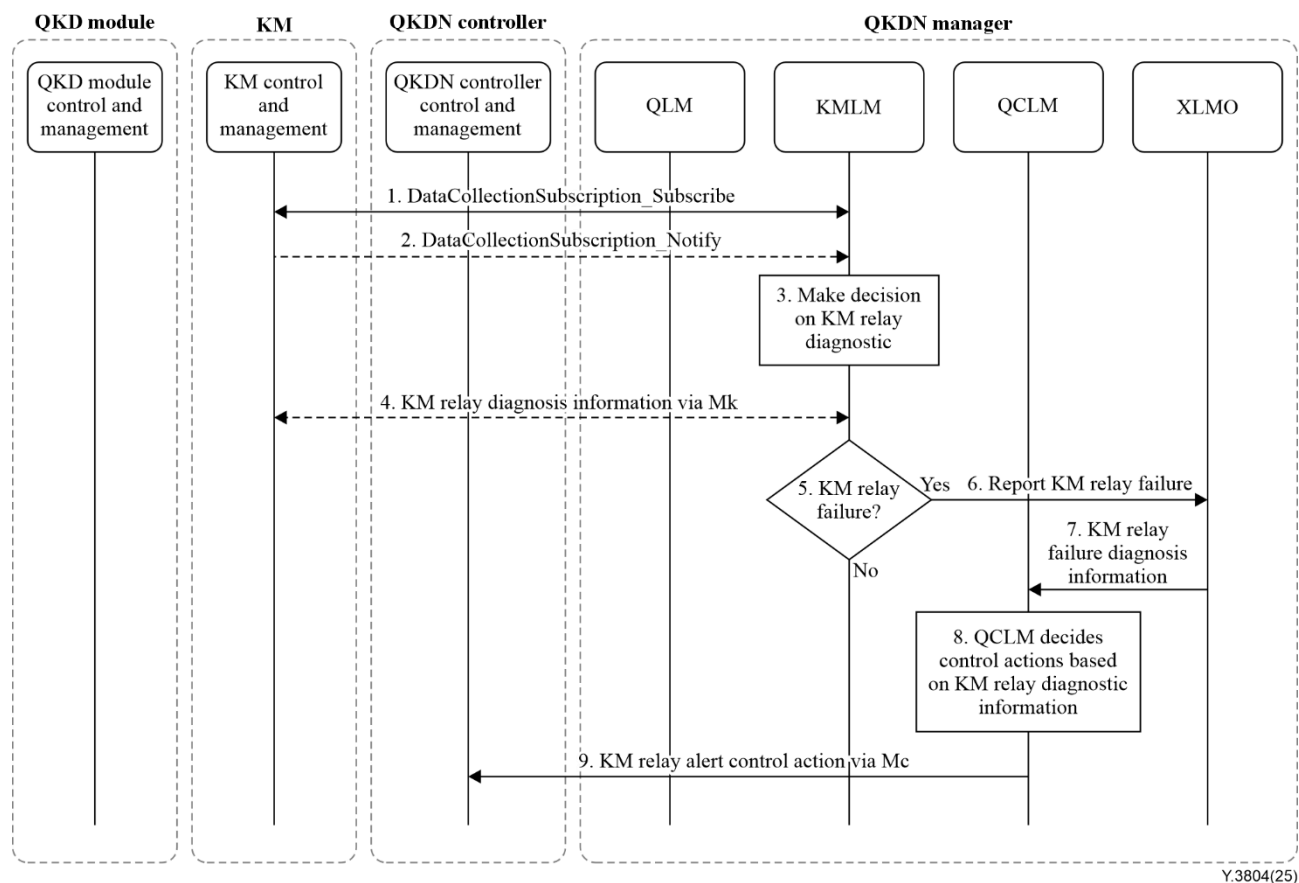


Figure 5 – A fault management procedure in subscription-notification information exchange: Key relay failure in KM

10.2 Accounting management procedure

- (1) Request-response information exchange

An accounting management procedure is shown in Figure 6.

- 1) A KMLM meters accounting information from the KM via Mk reference point.
- 2) The KMLM processes the metered accounting information and generates a report.
- 3) The KMLM sends the report to the XLMO for further charging process.
- 4) The XLMO then creates and stores charging data records (CDRs) based on the metered key management layer accounting reports and interacts with the billing system when it receives requests.

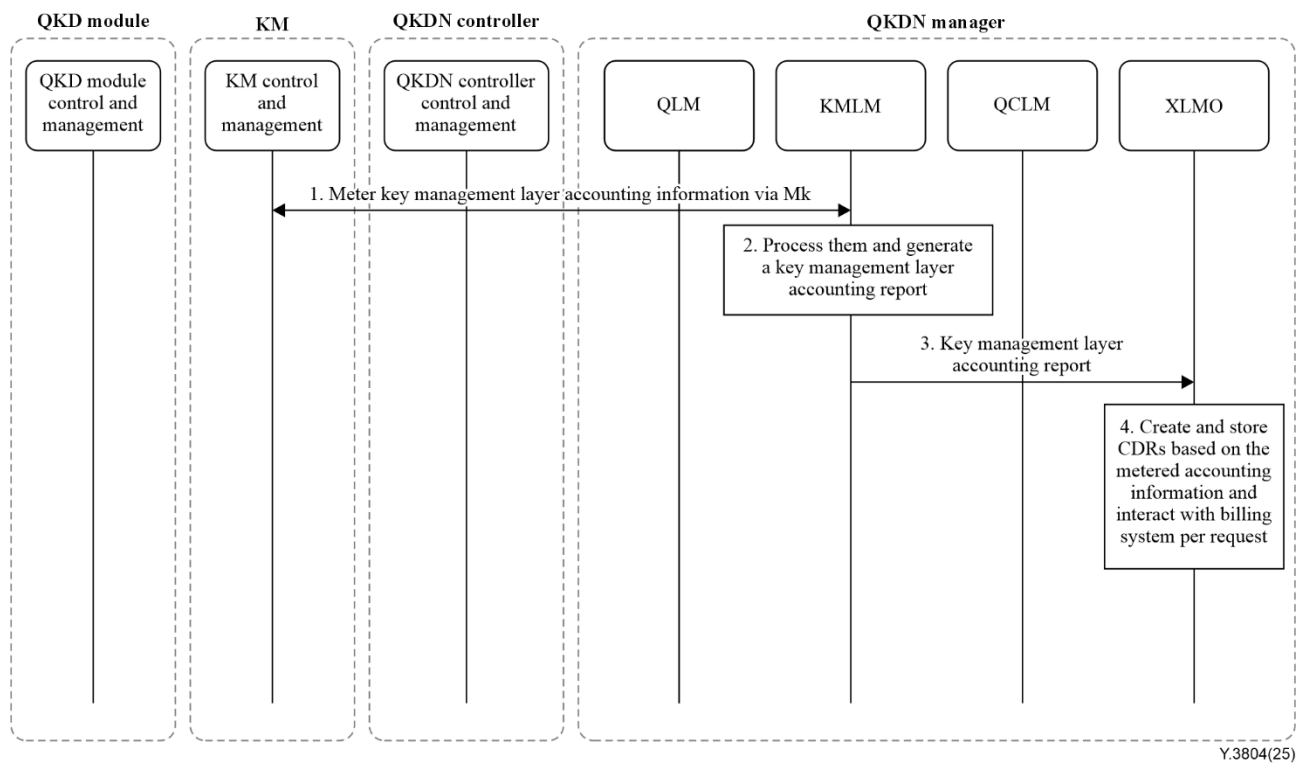


Figure 6 – A accounting management procedure in request-response information exchange

(2) Subscription-notification information exchange

An accounting management procedure is shown in Figure 7.

- 1) A KMLM subscribes to data collection by invoking the DataCollectionSubscription_Subscribe operation to the KM module control and management function via the Mk reference point. The KM module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the KMLM are listed in Table II.1. The TypeOperation indicates accounting management in this step.

- 2) The KM module control and management function notifies the KMLM with data collection by invoking DataCollectionSubscription_Notify operation, based on the subscription in step 1).

NOTE 2 – The information components that can be provided to the KMLM are listed in Table II.2. The DataCollectionSubscription_Notify includes an EventNotify. The EventNotify represents accounting information in this step.

- 3) The KMLM processes the metered accounting information and generates a report.
- 4) The KMLM sends the report to the XLMO for further charging process.
- 5) The XLMO, then, creates and stores charging data records (CDRs) based on the metered key management layer accounting reports and interacts with the billing system when it receives requests.

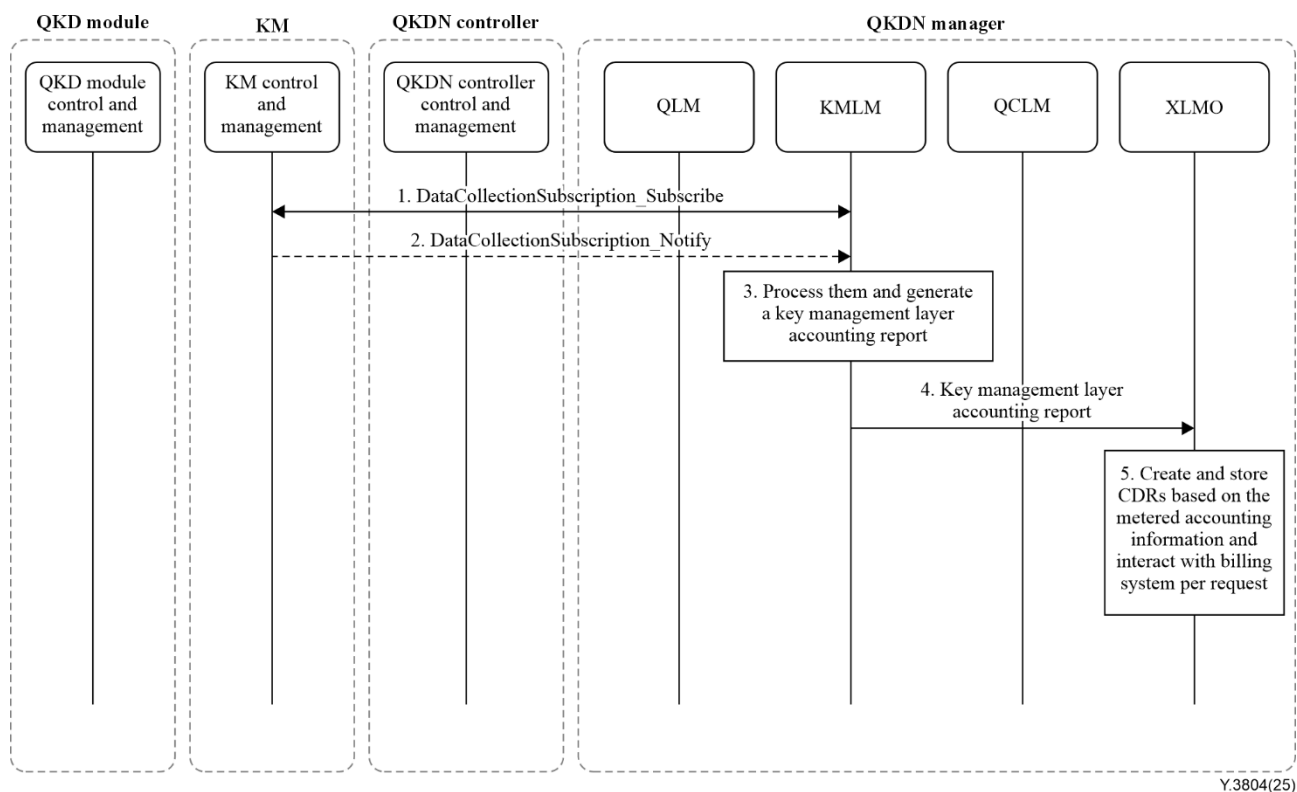


Figure 7 – An accounting management procedure in subscription-notification information exchange

10.3 Configuration management procedure

(1) Request-response information exchange

A configuration management procedure is shown in Figure 8.

- 1) A QLM discovers and collects quantum layer configuration information by the QKD module control and management function via the Mq reference point.
- 2) The QLM generates quantum layer topology and creates quantum layer resource inventory.
- 3) The QLM sends quantum layer configuration report to the XLMO.
- 4) The KMLM discovers and collects key management layer configuration information via Mk reference point.
- 5) The KMLM generates key management layer topology and creates key management layer resource inventory.
- 6) The KMLM sends key management layer configuration report to the XLMO.
- 7) The XLMO generates an entire QKDN topology by correlating quantum layer and key management layer topology information received, and also checks any configuration errors.

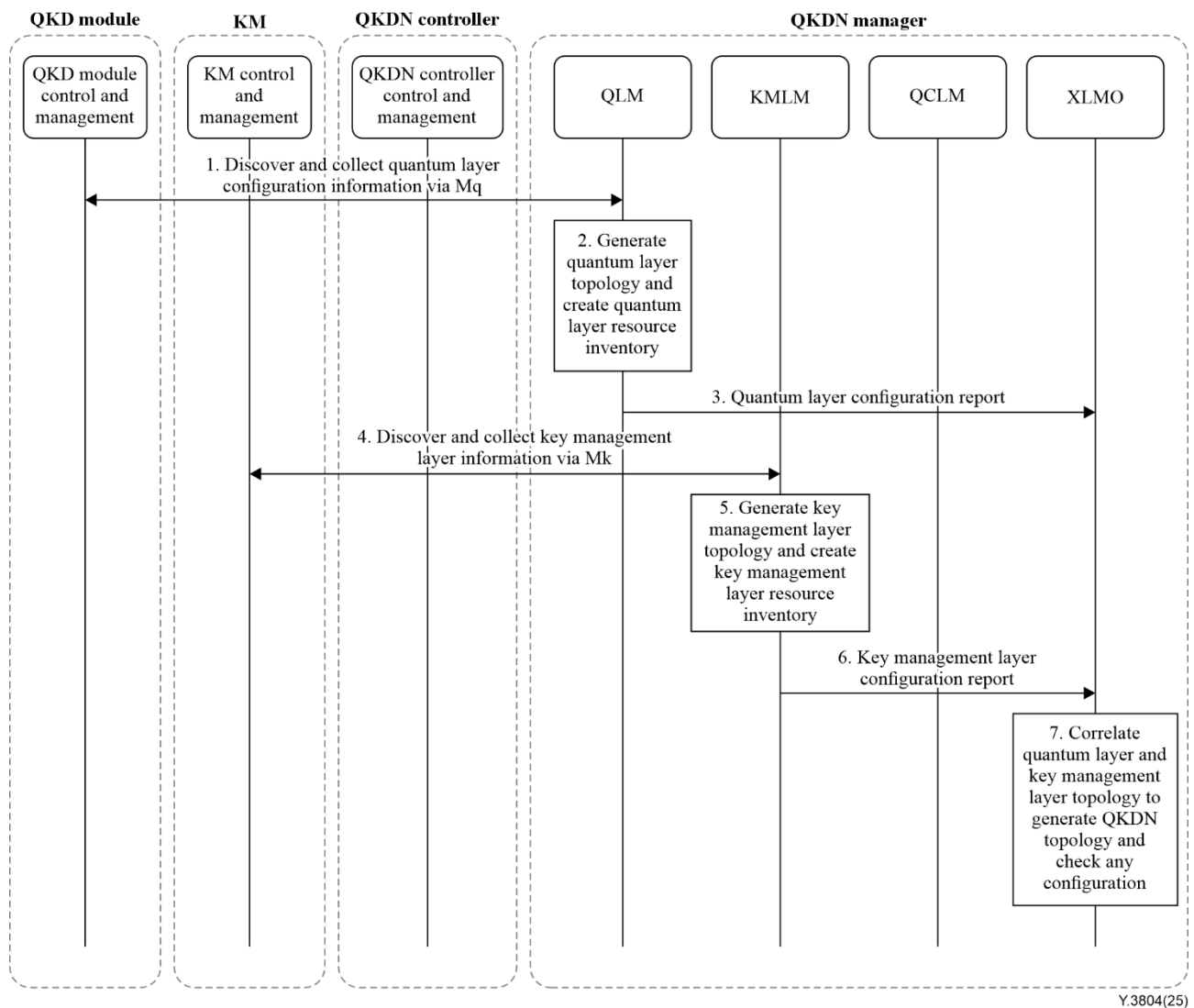


Figure 8 – A configuration management procedure in request-response information exchange

(2) Subscription-notification information exchange

A configuration management procedure is shown in Figure 9.

- 1) A QLM subscribes to data collection by invoking the DataCollectionSubscription_Subscribe operation to the QKD module control and management function via the Mq reference point. The QKD module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the QLM are listed in Table II.1. The TypeOperation indicates configuration management in this step.

- 2) The QKD module control and management function notifies the QLM with data collection by invoking DataCollectionSubscription_Notify operation, based on the subscription in step 1).

NOTE 2 – The information components that can be provided to the QLM are listed in Table II.2.

- 3) The QLM generates quantum layer topology and creates quantum layer resource inventory. QLM updates the analytics report including what have been done.

In steps 4) ~ 6), a KMLM also performs the same processes for key management layer configuration management as the QLM does in steps 1) ~ 3).

- 7) The QKD module control and management function subscribes to data analytics by invoking the AnalyticsSubscription_Subscribe operation to the QLM via the Mq reference point. The QLM replies with OK if the subscription is acceptable.

NOTE 3 – The information components that can be provided by the QKD module control and management function are listed in Table II.3. The TypeOperation indicates configuration management in this step.

- 8) The QLM notifies the QKD module control and management function with data analytics by invoking AnalyticsSubscription_Notify operation, based on the subscription in step 7).

NOTE 4 – The information components that can be provided to the QLM are listed in Table II.4.

In steps 9) ~ 10), the QLM also performs the same operations above for the XLMO, which depends on a decision of QLM.

In steps 11) ~ 14), the KMLM also performs the same processes for key management layer configuration management as the QLM does in steps 7) ~ 10).

- 15) The XLMO generates an entire QKDN topology by correlating quantum layer and key management layer topology information received and also checks any configuration errors.

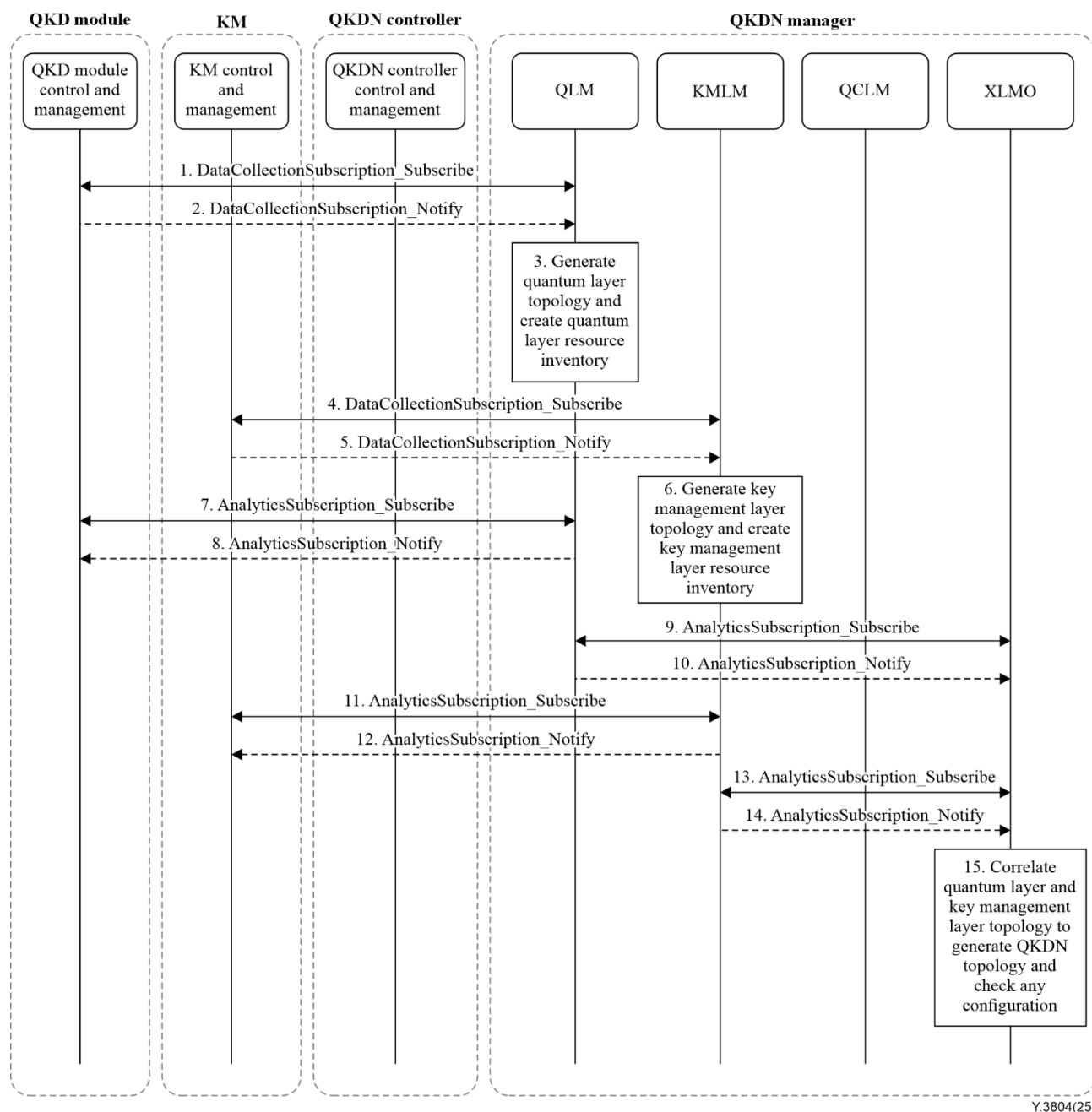


Figure 9 – A configuration management procedure in subscription-notification information exchange

10.4 Performance management procedure

(1) Request-response information exchange

A performance management procedure is shown in Figure 10.

- 1) A QLM collects quantum layer performance information by the QKD module control and management function via the Mq reference point.
- 2) The QLM performs analysis of the collected performance information and generates and stores a report.
- 3) The QLM then checks any performance degradation problems in the quantum layer. The QLM then sends the performance report to the XLMO.

In steps 4) ~ 6), a KMLM also performs the same processes for key management layer performance management as the QLM does in steps 1) ~ 3).

NOTE 1 – The order of collection process does not have any significance. Both can happen in parallel or in any order depending on the implementation.

- 7) The XLMO correlates the performance information of the quantum layer and key management layer performance to identify cross-layer performance degradation problems. It then creates a remedial action policy for both the quantum layer and the key management layer.
- 8) The XLMO then sends remedy control actions to the QCLM.
- 9) The QCLM sends it to the QKDN controller.
- 10) The QKDN controller forwards remedy actions to the control and management functions in the QKD module.
- 11) The QKDN controller also forwards them to the control and management functions in the KM.

NOTE 2 – Detailed suggestions for remedial decision making and associated actions are outside the scope of this Recommendation.

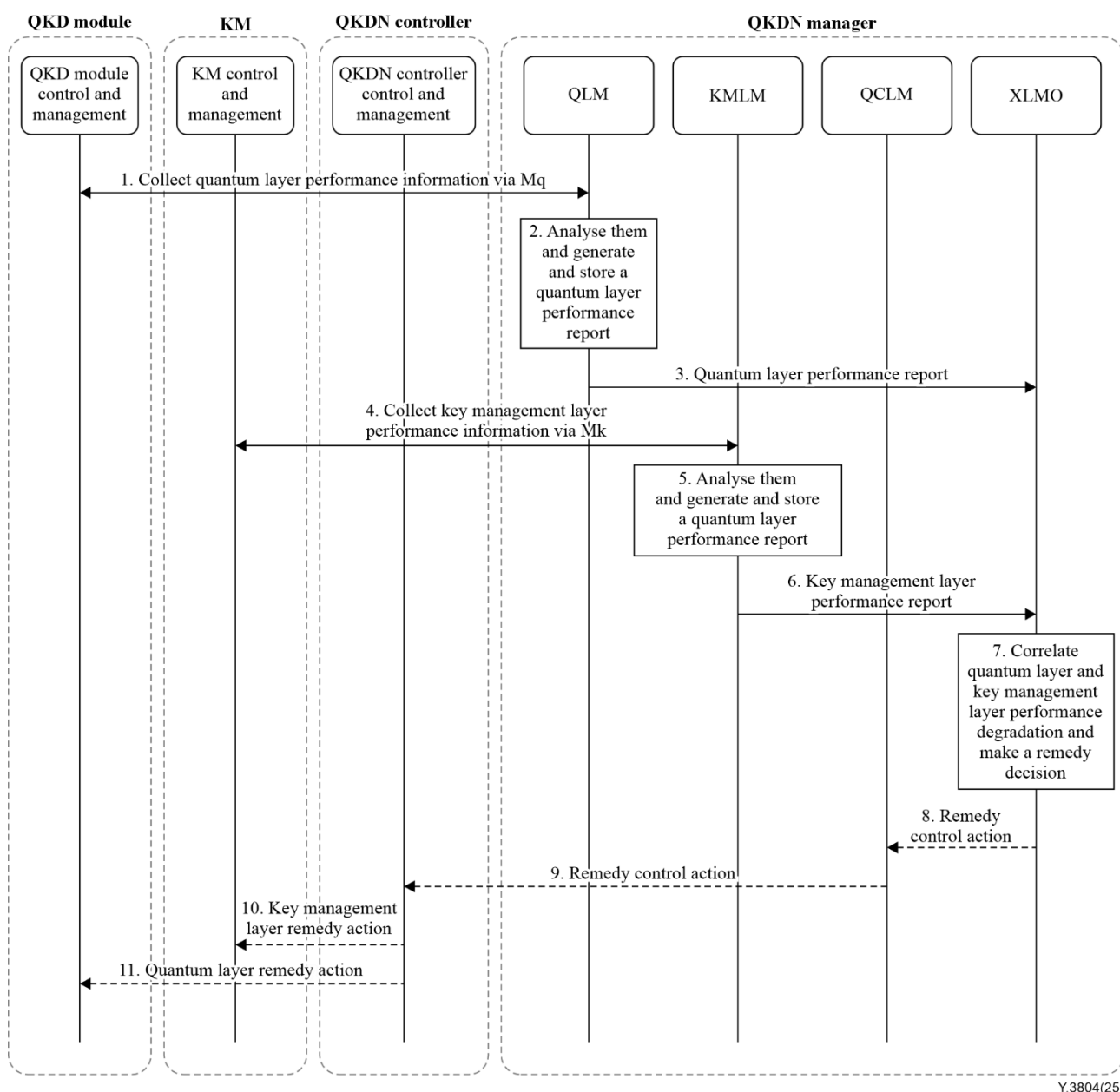


Figure 10 – A performance management procedure in request-response information exchange

(2) Subscription-notification information exchange

A performance management procedure is shown in Figure 11.

- 1) A QLM subscribes to data collection by invoking the DataCollectionSubscription_Subscribe operation to the QKD module control and management function via the Mq reference point. The QKD module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the QLM are listed in Table II.1. The TypeOperation indicates performance management in this step.

- 2) The QKD module control and management function notifies the QLM with data collection by invoking DataCollectionSubscription_Notify operation, based on the subscription in step 1).

NOTE 2 – The information components that can be provided to the QLM are listed in Table II.2.

- 3) The QLM performs analysis of the collected performance information and generates and stores an analytics report.
- 4) The QKD module control and management function subscribes to data analytics by invoking the AnalyticsSubscription_Subscribe operation to the QLM via the Mq reference point. The QLM replies with OK if the subscription is acceptable.

NOTE 3 – The information components that can be provided by the QKD module control and management function are listed in Table II.3. The TypeOperation indicates configuration management in this step.

- 5) The QLM notifies the QKD module control and management function with data analytics by invoking AnalyticsSubscription_Notify operation, based on the subscription in step 4).

NOTE 4 – The information components that can be provided to the QLM are listed in Table II.4.

- 6 ~ 7) The QLM also performs the same operations above for the XLMO, which depends on a decision of the QLM.

In steps 8) ~ 14), a KMLM also performs the same processes for key management layer performance management as the QLM does in steps 1) ~ 7).

NOTE 5 – The order of collection process does not have any significance. Both can happen in parallel or in any order depending on the implementation.

- 15) The XLMO correlates the performance information of the quantum layer and key management layer performance to identify cross-layer performance degradation problems. It then creates a remedial action policy for both the quantum layer and the key management layer.
- 16) The XLMO then sends remedy control actions to the QCLM
- 17) The QCLM sends it to the QKDN controller.
- 18) The QKDN controller forwards remedy actions to the control and management functions in the QKD
- 19) The QKDN controller also forwards them to the control and management functions in the KM.

NOTE 6 – Detailed suggestions for remedial decision making and associated actions are outside the scope of this Recommendation.

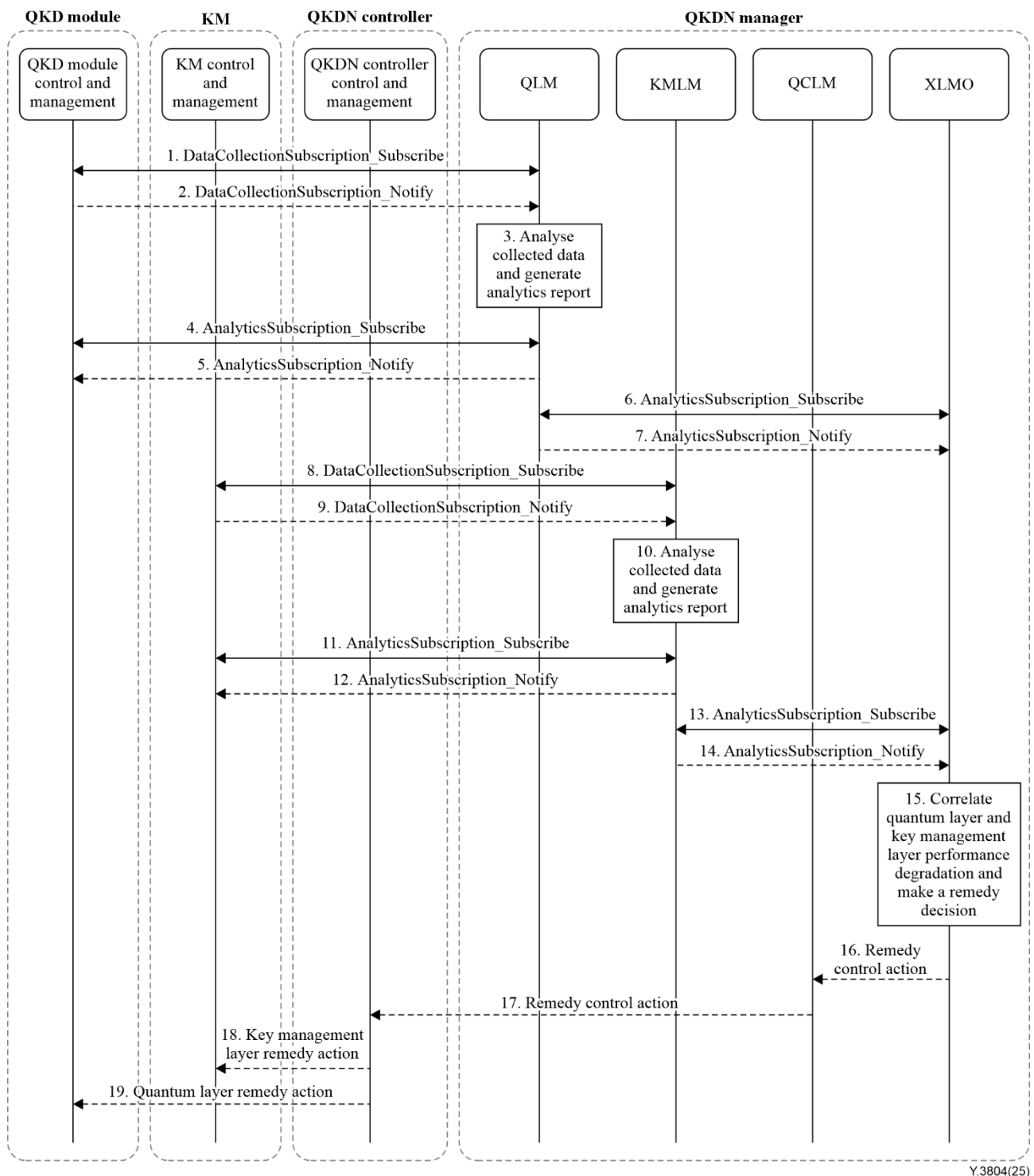


Figure 11 – A performance management procedure in subscription-notification information exchange

10.5 Security management procedure

(1) Request-response information exchange

A security management procedure is shown in Figure 12.

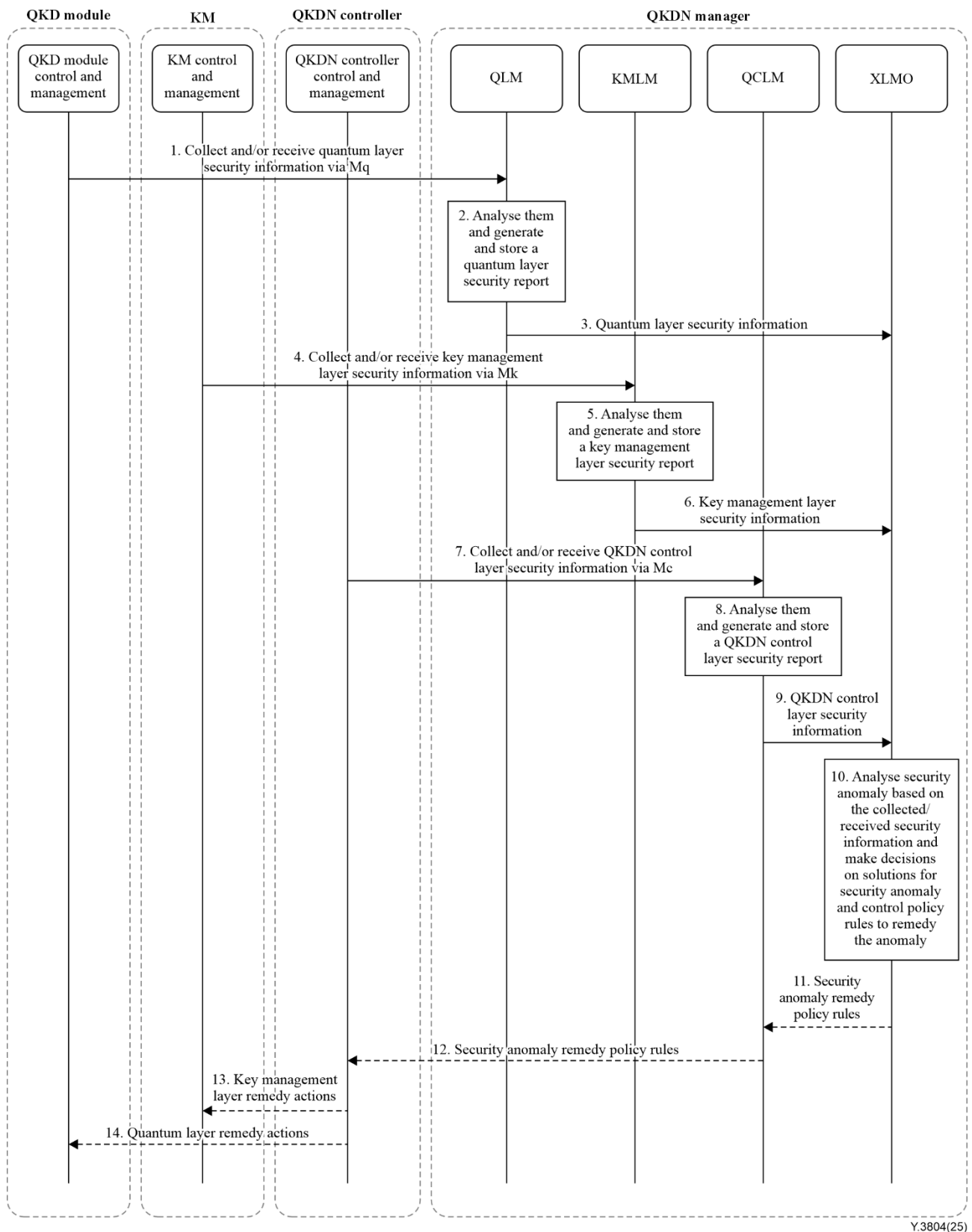
- 1) A QLM collects and/or receives the quantum layer security information by the QKD module control and management function via the Mq reference point.
- 2) The QLM analyses the collected security information and generates and stores a report.
- 3) The QLM then transfers quantum layer security information to the XLMO.

In steps 4) ~ 6) and steps 7) ~ 9) respectively, a KMLM and a QCLM also collect security information from the key management layer and the QKDN control layer as the QLM does in steps 1) ~ 3).

NOTE 1 – The order of collection of security information in layers may vary by implementation of QKDN.

- 10) The XLMO analyses the security anomaly based on the collected security information and makes a decision on solutions for security anomaly and control policy rules to remedy the anomaly.
- 11) The XLMO reports the resulting security anomaly remedy policy rules to the QCLM
- 12) The QCLM sends it to the QKDN controller.
- 13) The QKDN controller forwards remedy actions to the control and management functions in the QKD module.
- 14) The QKDN controller also forwards them to the control and management functions in the KM.

NOTE 2 – Detailed suggestions for remedial decision making, and associated actions are outside the scope of this Recommendation.



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Figure 12 – A security management procedure in request-response information exchange

(2) Subscription-notification information exchange

A security management procedure is shown in Figure 13.

- 1) A QLM subscribes to data collection by invoking the DataCollectionSubscription_Subscribe operation to the QKD module control and management function via the Mq reference point. The QKD module control and management function replies with OK if the subscription is acceptable.

NOTE 1 – The information components that can be provided by the QLM are listed in Table II.1. The TypeOperation indicates security management in this step.

- 2) The QKD module control and management function notifies the QLM with data collection by invoking DataCollectionSubscription_Notify operation, based on the subscription in step 1).

NOTE 2 – The information components that can be provided to the QLM are listed in Table II.2.

- 3) The QLM analyses the collected security information and generates and stores an analytics report.
- 4) A QKD module control and management function subscribes to data analytics by invoking the AnalyticsSubscription_Subscribe operation to the QLM via the Mq reference point. The QKD module control and management function replies with OK if the subscription is acceptable.

NOTE 3 – The information components that can be provided by the QKD module control and management function are listed in Table II.3. The TypeOperation indicates configuration management in this step.

- 5) The QLM notifies the QKD module control and management function with data analytics by invoking AnalyticsSubscription_Notify operation, based on the subscription in step 4).

NOTE 4 – The information components that can be provided to the QLM are listed in Table II.4.

- 6 ~ 7) The QLM also performs the same operations above for XLMO, which depends on a decision of QLM.

In steps 8) ~ 14) and 15) ~ 21) respectively, a KMLM and a QCLM also collect security information from the key management layer and the QKDN control layer, and the KMLM and the QCLM also report security analytics as the QLM does in steps 1) ~ 7).

NOTE 5 – The order of collection of security information in layers may vary by implementation of QKDN.

- 22) The XLMO analyses the security anomaly based on the collected security information and makes a decision on solutions for security anomaly and control policy rules to remedy the anomaly.
- 23) The XLMO reports the resulting security anomaly remedy policy rules to QCLM.
- 24) The QCLM sends it to the QKDN controller.
- 25) The QKDN controller forwards remedial actions to the control and management functions in the QKD module.
- 26) The QKDN controller also forwards them to the control and management functions in the KM.

NOTE 6 – Detailed suggestions for remedial decision making, and associated actions are outside the scope of this Recommendation.

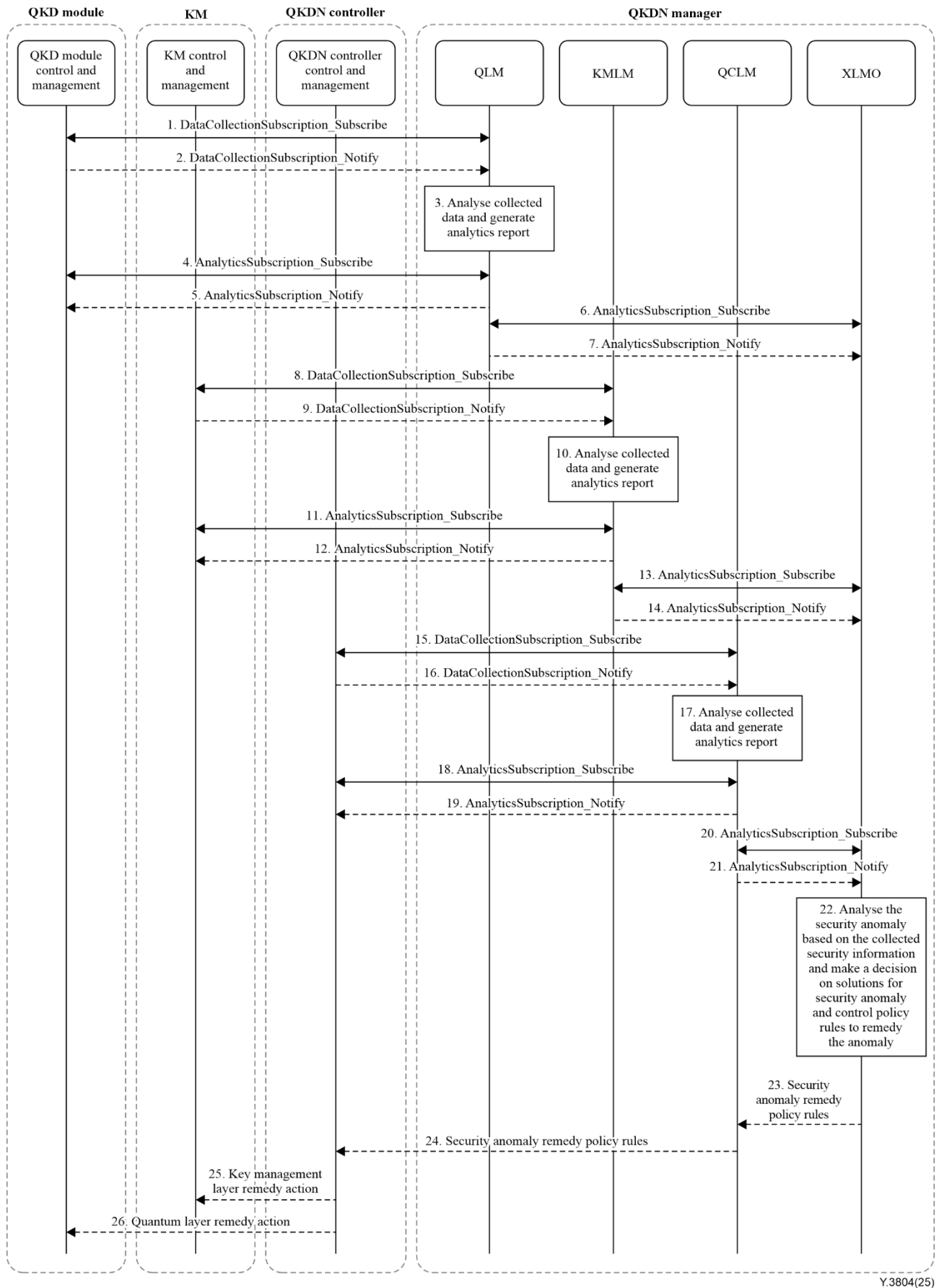


Figure 13 – A security management procedure in subscription-notification information exchange

10.6 Key relay procedure

A key relay control procedure is shown in Figure 14.

- 1) A KM requests a key relay route to the session control function in the QKDN controller.
- 2) The session control function checks whether there is any session between a source QKD node and a destination QKD node or not.
- 3) If there is no session, the session control function asks the routing control function for the key relay route.
- 4) The routing control function asks for the configuration control function configuration information.
- 5) The configuration control function replies with the configuration information.
- 6) The routing control function evaluates an optimal route based on the answered configuration information.
- 7) The routing control function replies with the route information to the session control function.
- 8) The session control function replies with the route information to the KM.

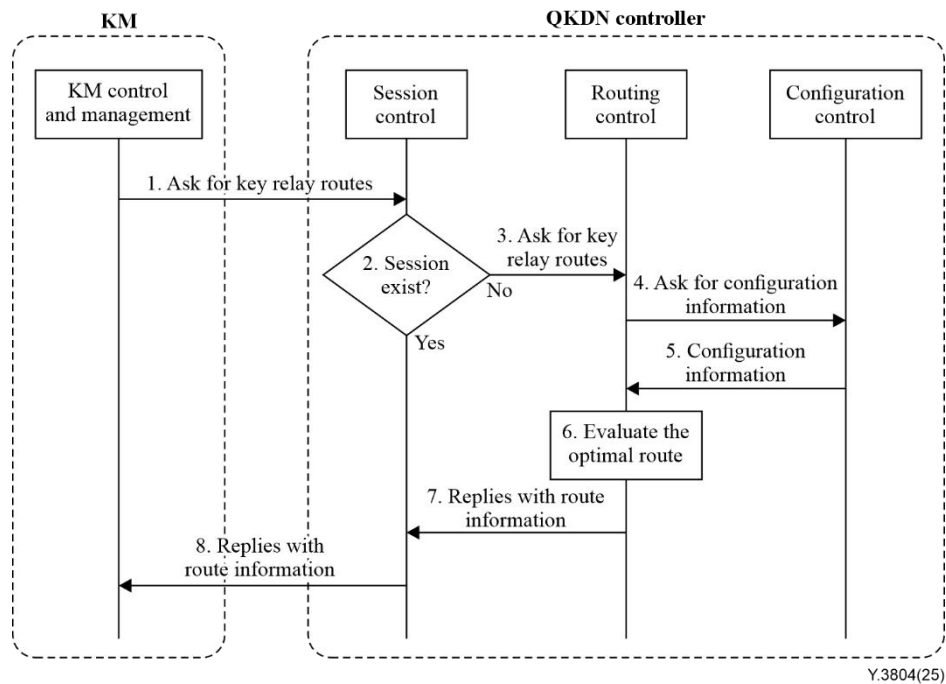


Figure 14 – A key relay procedure

10.7 Key relay rerouting procedure

A key relay rerouting procedure is shown in Figure 15.

- 1) A QCLM receives a key relay routing and re-routing support information requested by the QKDN controller control and management function.
- 2) The QCLM requests network topology information from the XLMO.
- 3) The XLMO reports it to the QCLM.
- 4) The QCLM analyses and selects matching key relay topology information and other supporting information.
- 5) The QCLM reports the resulting supporting information to the QKDN controller for controlling key relay routing and re-routing actions.

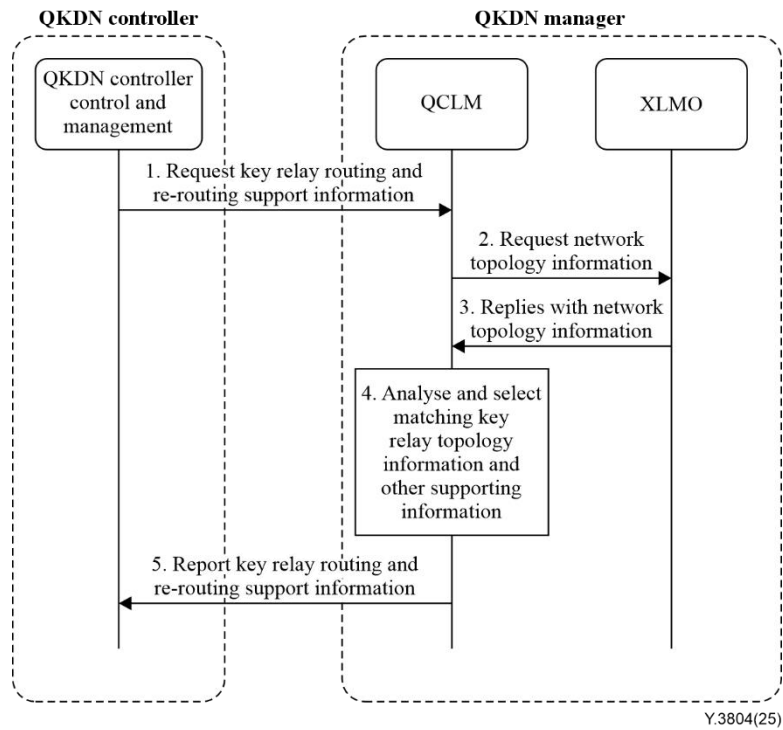


Figure 15 – A key relay rerouting procedure

11 Security considerations

In order to mitigate security threats and potential attacks, issues of confidentiality, integrity, authenticity, non-repudiation, availability, and traceability need to be addressed, and the appropriate security and privacy protection schemes should be considered in QKDN, the user network, and interfaces between the two networks. Details are outside the scope of this Recommendation.

Appendix I

Functions and information components for reference points

(This appendix does not form an integral part of this Recommendation.)

I.1 Reference points of the control, management and orchestration of QKDN

I.1.1 Session processing functions

To assure the reliability and performance of QKDN management session operations across a management reference point, the following capabilities are provided:

- **Overload control:** The reference point provides the function to support overload control for preventing the overflow of information messages exchanged.
- **Synchronization and audit:** The reference point is required to provide the function to support synchronization and audit of the QKDN management session status in support of recovery and operational information statistics and auditing.
- **Session state maintenance:** The reference point is required to be able to maintain the session state using either soft-state or hard-state approaches.

I.1.2 Information exchange functions

This clause provides a brief description of the information exchange functions for the reference point.

Request-response transactions: The reference point allows a QKDN manager (including QLM, KMLM, QCLM, and XLMO functions) to request a transaction to be performed by a responding QKDN functional component (including QKD link, QKD module, Key manager, and QKDN controller) and get a response (that can be correlated with the request) in return and also vice versa.

Subscription: The reference point supports the subscription for the notification between QKDN manager and QKDN functional components. In explicit subscription, a separate request-response exchange is performed. On the other hand, in implicit subscription, the subscription for notification is included as part of notification operation.

Notification: The reference point supports the notification of asynchronous events between QKDN manager and QKDN functional components.

Reliable delivery: The reference point provides reliable delivery of messages.

Capabilities: Each layer determines the capabilities of appropriate corresponding layer management functions.

Security: The reference point supports the authentication between two exchanging layers such that requests from unauthenticated sources will not be performed and such that each layer can verify the source of notifications sent.

One-to-many/many-to-one: Two modes are supported: 1) one-to-many mode: a QKDN management function communicates with multiple QKDN management functions; 2) many-to-one mode: multiple QKDN management functions make requests to a given QKDN management function.

I.2 Reference point common information components

This clause provides common information components applicable to all management reference points as shown in Table I.1.

Table I.1 – Common information components

Information component	Description
User identifier	A unique identifier for the layer management functional components (QCLM, KMLM, & QLM) within the same administrative domain of a single requestor.
Management operation request session identifier	An identifier for the session for which the management operation requests are sent to the QKDN layers (QKDN control, key management layer, and quantum layer). The identifier has to be unique within the same QKDN layer.
Globally unique IP address information (Optional)	A set of IP address information used for locating the network in which each layer control and management function is requesting the management operations.
Unique IP address	The IP address for identifying each layer control and management function.
Address realm	The addressing domain of the IP address (e.g., Subnet prefix or VPN ID)
Management operation requestor identifier	An identifier for the requestor of each layer management service. It is unique over the requestors sending requests for each layer management functional component (QCLM, KMLM, & QLM).
Management operation request priority (Optional)	The indication of the importance of a management operation request. It can be used for processing simultaneous requests by each layer management functional component (QCLM, KMLM, & QLM) based on the priority level.
Management operation request result	Indication of the result for a management operation request (includes both synchronous and scheduled request result).
EventNotify	Allows each layer to send notifications to the layer specific management function in QKDN manager for an event that may need to take an appropriate action for requested management operations.

I.3 Reference point Mc

The Mc reference point is required to support request-response information exchange and subscription-notification information exchange between the QCLM function of QKDN manager and the QKDN controller control and management function in QKDN control layer.

The Mc reference point operates as an intra-domain reference point.

I.3.1 Mc functions for QCLM

The Mc reference point provides the following functions for QCLM:

- Provisioning of QKDN control layer functions;
- Monitoring performance of QKDN control layer functions;
- Collecting alarm information of QKDN control layer functions;
- Requesting management decision for control actions (e.g., path configuration information, etc.);
- Requesting status of network operations (e.g., routing status, etc.);

- Performing control orchestration actions (e.g., path modification for load-balancing, etc.).

I.3.2 Information components

The information components exchanged across the Mc reference point are categorized in Table I.2.

Table I.2 – Information components for reference point Mc

Information component	Description
Control orchestration action description	Describes information components related with control orchestration actions
Control orchestration action type	The indication of types of control orchestration action (e.g., path modification, path re-provisioning, etc.)
Control orchestration action priority	The indication of the priority of the action
Control orchestration action value	Specification of the control orchestration action per action type

I.4 Reference point Mk

The Mk reference point supports request-response information exchange and subscription-notification information exchange between the KMLM function of QKDN manager and the KM control and management function in the QKDN key management layer.

The Mk reference point operates as an intra-domain reference point.

I.4.1 Mk functions for KMLM

The Mk reference point provides the following functions for KMLM:

- Provisioning of key management layer functions;
- Monitoring performance of key management layer functions;
- Collecting alarm information of key management layer functions (e.g., key generation failure;
- Collecting key life cycle management information, QKD link parameter status, etc.

I.4.2 Information components

The information components exchanged across the Mk reference point are categorized in Table I.3.

Table I.3 – Information components for reference point Mk

Information component	Description
Key management layer resource description	Description of key management layer resource for management purpose
Key management layer managed resource ID	Identifier of key management layer managed resource of interests for management
Key management layer performance description	Description of performance information of the key management layer managed resource
Key management layer performance information type	Indication of type of performance information (e.g., key relaying delay, key data loss, etc.)
Key management layer performance information target managed object ID	Identifier of target managed object for collecting performance information

Table I.3 – Information components for reference point Mk

Information component	Description
Key management layer performance information value	Value of the performance information collected
Key Management Layer Alarm Description	Description of alarm information of the key management layer managed resource
Key management layer alarm type	Indication of type of key management layer alarm associated with fault or anomalous event
Key management layer alarm ID	Identifier of a key management layer alarm associated with fault or anomalous event
Key management layer alarm source ID	Identifier of a source of a key management layer alarm
Key management layer alarm value	Value of the key management layer alarm

I.5 Reference point Mq

The Mq reference point supports request-response information exchange and subscription-notification information exchange between the QLM function of QKDN manager and the QKD module control and management function in QKDN quantum layer.

The Mq reference point operates as an intra-domain reference point.

I.5.1 Mq functions for QLM

The Mq reference point provides the following functions for QLM:

- Provisioning of QKD module functions in the quantum layer;
- Monitoring performance of QKD module functions in the quantum layer;
- Collecting alarm information of QKD module functions in the quantum layer.

I.5.2 Information components

The information components exchanged across the Mq reference point are categorized in Table I.4.

Table I.4 – Information components for reference point Mq

Information component	Description
QKD module in quantum layer resource description	Description of QKD module in quantum layer resource for management purpose
QKD module in quantum layer managed resource ID	Identifier of QKD module in quantum layer managed resource of interests for management
QKD module in quantum layer performance description	Description of performance information of the QKD module in quantum layer managed resource
QKD module in quantum layer performance information type	Indication of type of performance information (e.g., key relaying delay, key data loss, etc.)
QKD module in quantum layer performance information target managed object ID	Identifier of target managed object for collecting performance information
QKD module in quantum layer performance information value	Value of the performance information collected
QKD module in quantum layer alarm description	Description of alarm information of the QKD module in quantum layer managed resource

Table I.4 – Information components for reference point Mq

Information component	Description
QKD module in quantum layer alarm type	Indication of type of QKD module in quantum layer alarm associated with fault or anomalous event
QKD module in quantum layer alarm ID	Identifier of a QKD module in quantum layer alarm associated with fault or anomalous event
QKD module in quantum layer alarm source ID	Identifier of a source of a QKD module in quantum layer alarm
QKD module in quantum layer alarm value	Value of the QKD module in quantum layer alarm

I.6 Reference point Mops

The Mops reference point supports request-response information exchange and subscription-notification information exchange between the QLM function of QKDN manager and the optical switching/splitting function in QKD link of the QKDN quantum layer.

The Mops reference point operates as an intra-domain reference point.

I.6.1 Mops functions for QLM

The Mops reference point provides the following functions for QLM:

- Provisioning of QKD-link functions in the quantum layer;
- Monitoring the performance of QKD-link functions in the quantum layer;
- Collecting alarm information of QKD-link functions in the quantum layer.

I.6.2 Information components

The information components exchanged across the Mops reference point are categorized in Table I.5.

Table I.5 – Information components for reference point Mops

Information component	Description
QKD link in quantum layer resource description	Description of QKD link in quantum layer resource for management purpose
QKD link in quantum layer managed resource ID	Identifier of QKD link in quantum layer managed resource of interests for management
QKD link in quantum layer performance description	Description of performance information of the QKD link in quantum layer managed resource
QKD link in quantum layer performance information type	Indication of type of performance information (e.g., key relaying delay, key data loss, etc.)
QKD link in quantum layer performance information target managed object ID	Identifier of target managed object for collecting performance information
QKD link in quantum layer performance information value	Value of the performance information collected
QKD link in quantum layer alarm description	Description of alarm information of the QKD link in quantum layer managed resource
QKD link in quantum layer alarm type	Indication of type of QKD link in quantum layer alarm associated with fault or anomalous event
QKD link in quantum layer alarm ID	Identifier of a QKD link in quantum layer alarm associated with fault or anomalous event

Table I.5 – Information components for reference point Mops

Information component	Description
QKD link in quantum layer alarm source ID	Identifier of a source of a QKD link in quantum layer alarm
QKD link in quantum layer alarm value	Value of the QKD link in quantum layer alarm

I.7 Reference point Mu

The Mu reference point supports request-response between the cross-layer management orchestration (XLMO) function of QKDN manager and the user network manager in user network management layer.

The Mu reference point operates as an intra-domain reference point.

I.7.1 Functions for XLMO

The Mu reference point provides the following functions for XLMO:

- Requesting provisioning status of user network management layer functions;
- Requesting performance of user network management layer functions;
- Requesting alarm information of user network management layer functions (e.g., user network node/link failure);
- Responding and/or sending asynchronous notifications to QKDN management operation requests from user network manager(s);
- Requesting-responding to QKDN management operations to any other external management entities besides user network manager(s) if any.

I.7.2 Information components

The information components exchanged across the Mu reference point are categorized in Table I.6.

Table I.6 – Information components for reference point Mu

Information component	Description
User network management layer resource Description	Description of user network management layer resource for management purpose
User network management layer managed resource ID	Identifier of user network management layer managed resource of interests for management
User network management layer Performance Description	Description of performance information of the user network management layer managed resource
User network management layer performance information type	Indication of type of performance information (e.g., key relaying delay, key data loss, etc.)
User network management layer performance information target managed object ID	Identifier of target managed object for collecting performance information
User network management layer performance information value	Value of the performance information collected
User network management layer Alarm Description	Description of alarm information of the user network management layer managed resource
User network management layer alarm type	Indication of type of user network management layer alarm associated with fault or anomalous event

Table I.6 – Information components for reference point Mu

Information component	Description
User network management layer alarm ID	Identifier of a user network management layer alarm associated with fault or anomalous event
User network management layer alarm source ID	Identifier of a source of a user network management layer alarm
User network management layer alarm value	Value of the user network management layer alarm

I.8 Reference point Mx

The Mx reference point supports request-response information exchange and subscription-notification information exchange between the XLMO function of two QKDN managers.

The Mx reference point operates as an intra-domain reference point.

I.8.1 Functions for XLMO

The Mx reference point provides the following functions for XLMO:

- Requesting provisioning status of QKDN management layer functions;
- Requesting performance of QKDN management layer functions;
- Requesting alarm information of QKDN management layer functions;
- Responding and/or sending asynchronous notifications to QKDN management operation requests from the other QKDN manager(s);
- Requesting-responding to QKDN management operations to any other external management entities besides the other QKDN manager(s) if any.

I.8.2 Information components

The information components exchanged across the Mx reference point are categorized in Table I.7.

Table I.7 – Information components for reference point Mx

Information component	Description
QKDN management layer resource Description	Description of QKDN management layer resource for management purpose
QKDN management layer managed resource ID	Identifier of QKDN management layer managed resource of interests for management
QKDN management layer Performance Description	Description of performance information of the QKDN management layer managed resource
QKDN management layer performance information type	Indication of type of performance information (e.g., key relaying delay, key data loss, etc.)
QKDN management layer performance information target managed object ID	Identifier of target managed object for collecting performance information
QKDN management layer performance information value	Value of the performance information collected
QKDN management layer Alarm Description	Description of alarm information of the QKDN management layer managed resource
QKDN management layer alarm type	Indication of type of QKDN management layer alarm associated with fault or anomalous event

Table I.7 – Information components for reference point Mx

Information component	Description
QKDN management layer alarm ID	Identifier of a QKDN management layer alarm associated with fault or anomalous event
QKDN management layer alarm source ID	Identifier of a source of a QKDN management layer alarm
QKDN management layer alarm value	Value of the QKDN management layer alarm

Appendix II

Information components for Subscription-Notification information exchange

(This appendix does not form an integral part of this Recommendation.)

II.1 Data collection operation

This clause provides information components for data collection operations. Table II.1 shows information components applicable to DataCollectionSubscription_Subscribe.

Table II.1 – Information components for DataCollectionSubscription_Subscribe

Information component	Description	Value
Consumer identifier	A unique identifier for a function of requesting management operation.	QCLM, KMLM, QLM
Producer identifier	A unique identifier for a function of responding management operation.	QKDN controller, KM, QKD module
Globally unique IP address information (Optional)	A set of IP address information used for locating the network in which each layer control and management function is requesting the management operations.	QKDN manager, QKDN controller, KM, QKD module
Unique IP address	The IP address for identifying each layer control and management function.	QCLM, KMLM, QLM
Address realm	The addressing domain of the IP address (e.g., Subnet prefix or VPN ID)	
Management operation layer (Optional)	A layer for operating performance management	QKDN control layer, key management layer, quantum layer
Management operation request priority (Optional)	The indication of the importance of a management operation request. It can be used for processing simultaneous requests by each layer management functional component (QCLM, KMLM, & QLM) based on the priority level.	Priority level
TypeNotify (Optional)	The indication of the type of notifying an event.	Time interval, Periodicity, Maximum number, Minimum number
TypeOperation	The indication of the type of the management operation.	Fault, Accounting, Configuration, Performance, Security
Correlation Identifier	A unique identifier for correlation between a consumer and a producer.	
EventFilter	The indication of the conditions to be fulfilled for reporting Event Information. This provides a set of Event parameter types and values, and enables to select which type of Event information is requested.	Event Filter information

Table II.2 shows information components applicable to DataCollectionSubscription_Notify.

Table II.2 – Information components for DataCollectionSubscription_Notify

Information component	Description	Value
Consumer identifier	A unique identifier for a function of requesting management operation.	QCLM, KMLM, QLM
Producer identifier	A unique identifier for a function of responding management operation.	QKDN controller, KM, QKD module
Globally unique IP address information (Optional)	A set of IP address information used for locating the network in which each layer control and management function is requesting the management operations.	QKDN manager, QKDN controller, KM, QKD module
Unique IP address	The IP address for identifying each layer control and management function.	QCLM, KMLM, QLM
Address realm	The addressing domain of the IP address (e.g., Subnet prefix or VPN ID)	
Management operation layer (Optional)	A layer for operating performance management	QKDN control layer, key management layer, quantum layer
Management operation request priority (Optional)	The indication of the importance of a management operation request. It can be used for processing simultaneous requests by each layer management functional component (QCLM, KMLM, & QLM) based on the priority level.	Priority level
Correlation Identifier	A unique identifier for correlation between a consumer and a producer.	
EventReport	The Event Report Information for the requested Event Filter information.	Event Report information

II.2 Analytics operation

This clause provides information components for data analytics operations. Table II.3 shows information components applicable to AnalyticsSubscription_Subscribe.

Table II.3 – Information components for AnalyticsSubscription_Subscribe

Information component	Description	Value
Consumer identifier	A unique identifier for a function of requesting performance management operation.	QKDN controller, KM, QKD module
Producer identifier	A unique identifier for a function of responding performance management operation.	QCLM, KMLM, QLM
Globally unique IP address information (Optional)	A set of IP address information used for locating the network in which each layer control and management function is requesting the management operations.	QKDN manager, QKDN controller, KM, QKD module
Unique IP address	The IP address for identifying each layer control and management function.	QCLM, KMLM, QLM
Address realm	The addressing domain of the IP address (e.g., Subnet prefix or VPN ID)	
Management operation layer (Optional)	A layer for operating performance management	QKDN control layer, key management layer, quantum layer

Table II.3 – Information components for AnalyticsSubscription_Subscribe

Information component	Description	Value
Management operation request priority (Optional)	The indication of the importance of a management operation request. It can be used for processing simultaneous requests by each layer management functional component (QCLM, KMLM, & QLM) based on the priority level.	Priority level
TypeNotify (Optional)	The indication of the type of notifying an event.	Time interval, Periodicity, Maximum number, Minimum number
TypeOperation	The indication of the type of the management operation.	Fault, Accounting, Configuration, Performance, Security
Correlation Identifier	A unique identifier for correlation between a consumer and a producer.	
AnalyticsFilter	The indication of the conditions to be fulfilled for reporting Analytics Information. This provides a set of Analytics parameter types and values, and enables to select which type of Analytics information is requested.	Analytics Filter information

Table II.4 shows information components applicable to AnalyticsSubscription_Notify.

Table II.4 – Information components for AnalyticsSubscription_Notify

Information component	Description	Value
Consumer identifier	A unique identifier for a function of requesting management operation.	QKDN controller, KM, QKD module
Producer identifier	A unique identifier for a function of responding management operation.	QCLM, KMLM, QLM
Globally unique IP address information (Optional)	A set of IP address information used for locating the network in which each layer control and management function is requesting the management operations.	QKDN manager, QKDN controller, KM, QKD module
Unique IP address	The IP address for identifying each layer control and management function.	QCLM, KMLM, QLM
Address realm	The addressing domain of the IP address (e.g., Subnet prefix or VPN ID)	
Management operation layer (Optional)	A layer for operating performance management	QKDN control layer, key management layer, quantum layer

Table II.4 – Information components for AnalyticsSubscription_Notify

Information component	Description	Value
Management operation request priority (Optional)	The indication of the importance of a management operation request. It can be used for processing simultaneous requests by each layer management functional component (QCLM, KMLM, & QLM) based on the priority level.	Priority level
Correlation Identifier	A unique identifier for correlation between a consumer and a producer.	
AnalyticsReport	The Analytics Report Information for requested Analytics Filter information.	Analytics Report information

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